

From Weakest Link to Strongest Asset

Building a Human Resilience Culture in Education

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apparentsecurity





"Just because they're aware doesn't mean they care."

— Perry Carpenter



30%

of breaches involve phishing.

Source: 2016 Verizon Data Breach Investigations Report



68%

of breaches involve the human element.

Source: 2026 Verizon Data Breach Investigations Report

whoami



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I figured out Forrest Gump's password





1-Forrest-1

Bridging the Gap Between Awareness & Resilience

Security Awareness Gap

A Hero's Journey

Homework


Bank
System Alert
Username
Password
Update
Vulnerability Exploit
Email
Password Expiry
Password Expiry
System
Impact


Bank
System Alert
Update
AI-POWERED PHISHING
Deepfake voice call to confirm email, adaptive template
AI-AUTOMATED EXPLOIT
Vulnerability discovery & zero-day prediction
LLM-GENERATED ADVERSARIAL SAMPLES
Evasion of next-gen AV
Password Expiry
GENAI CREDENTIAL HARVESTING
Context-aware email tailored to cite
ADVERSARIAL ATTACK AI
Graph anomaly detection vs adaptive evasion
VS
AI-POWERED DEFENSE MODEL
Graph anomaly detection vs adaptive evasion
AI-OPTIMIZED EXFILTRATION

The Old Equation

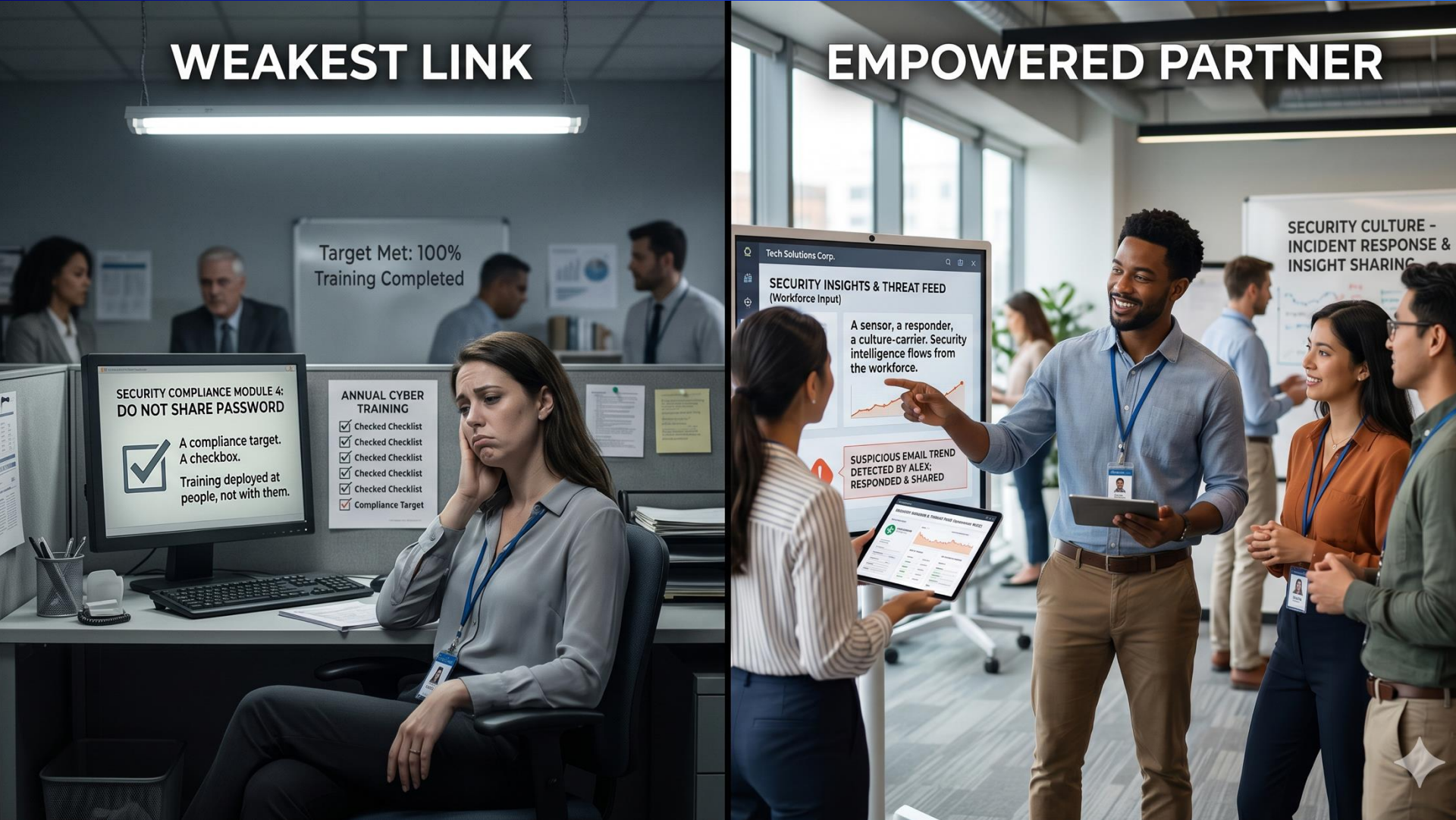
Phishing Training
+ Annual Click Rates

"Security Awareness"

Strengthening Your

HUMAN RESILIENCE

is about the intersection between ***Awareness***, ***Intention***, and ***Behaviors***


WEAKEST LINK
Target Met: 100%
Training Completed
SECURITY COMPLIANCE MODULE 4:
DO NOT SHARE PASSWORD
A compliance target.
A checkbox.
Training deployed at people, not with them.
ANNUAL CYBER TRAINING
Checked Checklist
Checked Checklist
Checked Checklist
Checked Checklist
Checked Checklist
Compliance Target
EMPOWERED PARTNER
SECURITY CULTURE - INCIDENT RESPONSE & INSIGHT SHARING
Tech Solutions Corp.
SECURITY INSIGHTS & THREAT FEED
(Workforce Input)
A sensor, a responder, a culture-carrier. Security intelligence flows from the workforce.
SUSPICIOUS EMAIL TREND DETECTED BY ALEX; RESPONDED & SHARED

Security Awareness and Human Resilience – the GAP



Awareness is not behavior. Behavior is not culture. Culture is what most programs never reach.

Why Organizations Need Human Resilience Programs



Humans Are the Primary Attack Vector



Traditional SAT Is Insufficient



Risk Reduction Is Measurable



Cost of Inaction Is High



Insider Risk and Negligence



Regulatory and Governance Pressure



Security Culture is a Strategic Asset



Culture doesn't come from policy documents.



It comes from what leaders model under pressure.

Two Organizations, Same Attack

An attacker sent the same email to two institutions.

Same template.

Same urgency.

Same spoofed domain.

What happened next had nothing to do with technology.

Santa Security Training

Wire transferred.

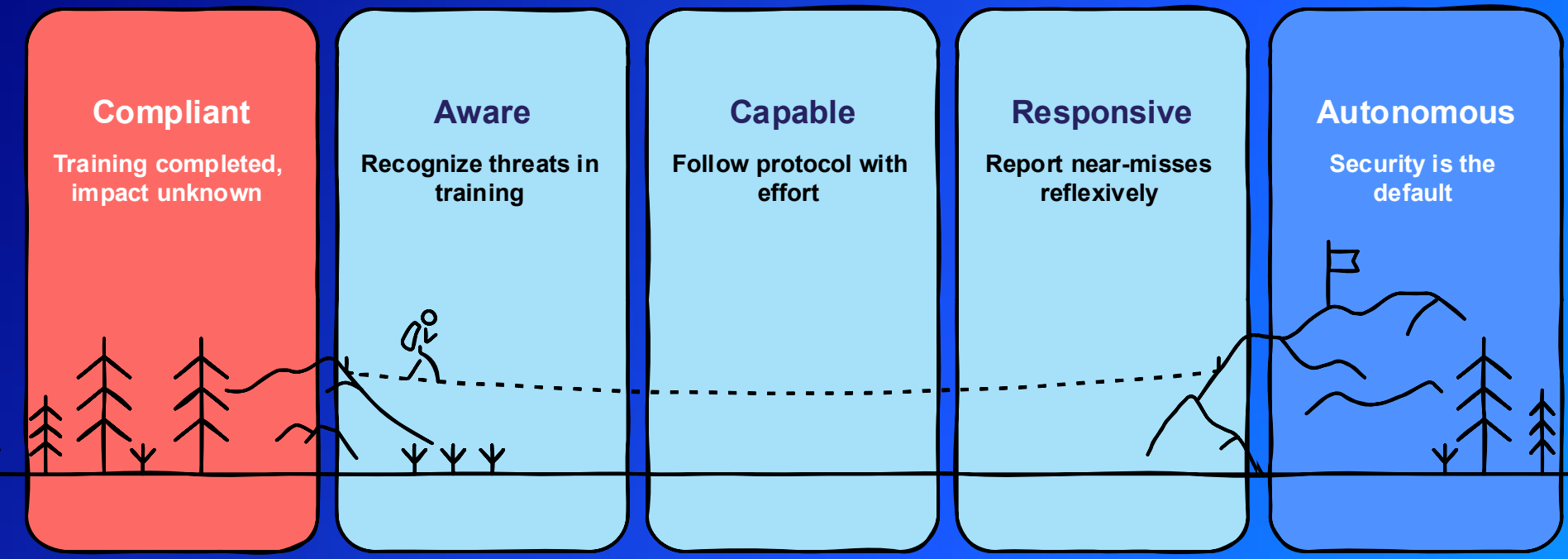
- Discovered 3 days later
- FERPA notification required
- Emergency board meeting called
- Vendor relationship under legal review
- $47,000 unrecovered
- Ongoing Weeks later

Human Resilient Culture

Attack chain broken.

- Staff paused as something felt wrong
- Called vendor directly
- Reported to security within minutes
- Full incident documented in minutes
- No breach. No meeting. No loss.
- Resolved <20 minutes

Where Do you Stand With Your Users or Program?



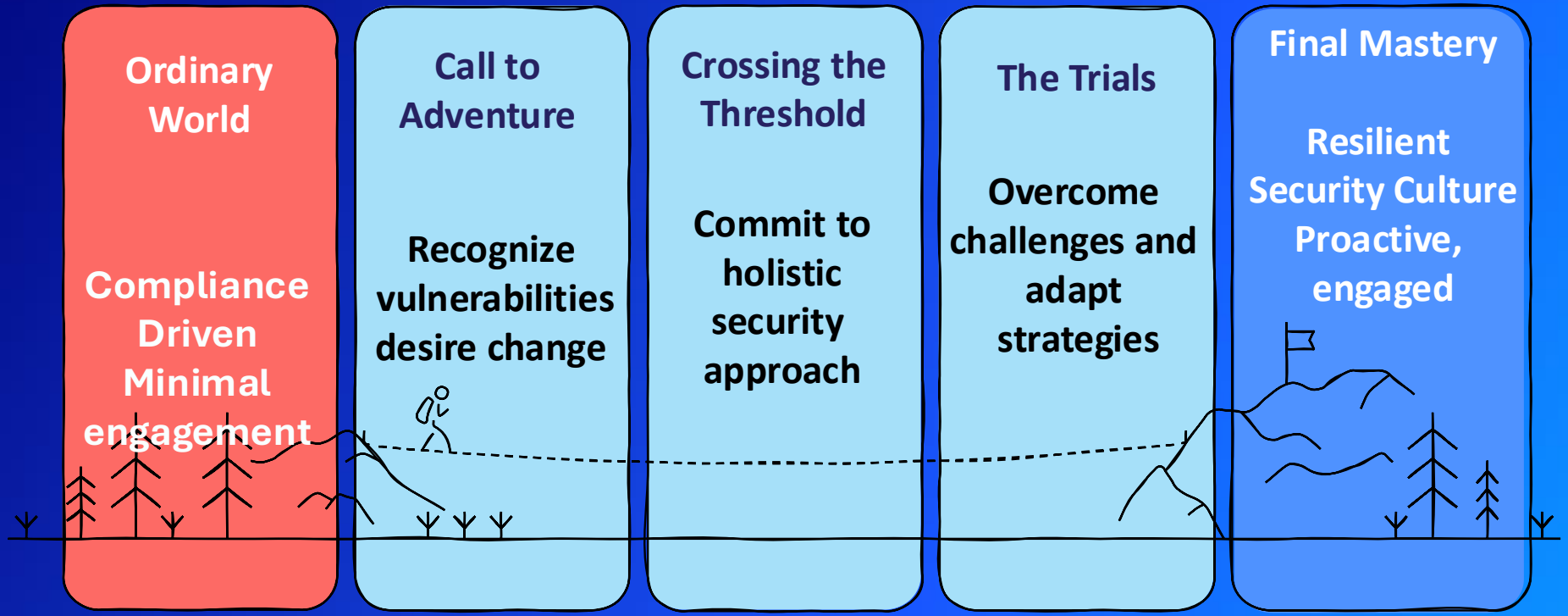


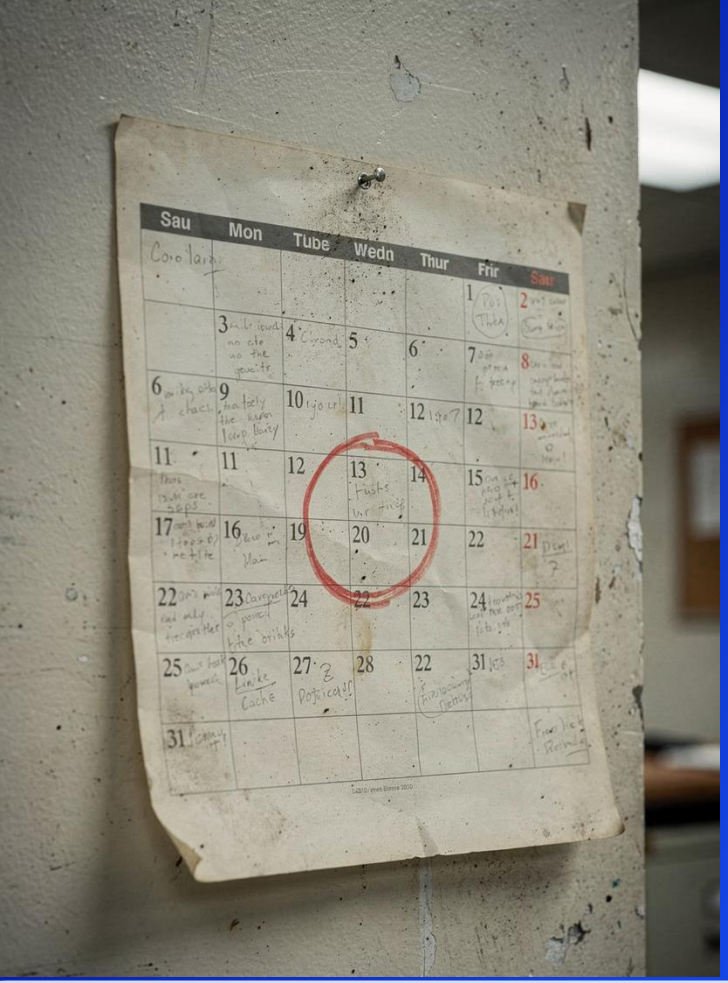
A Hero’s Journey





The Hero's Journey





The Ordinary World

The training happened.
Nobody knows if
anything changed.

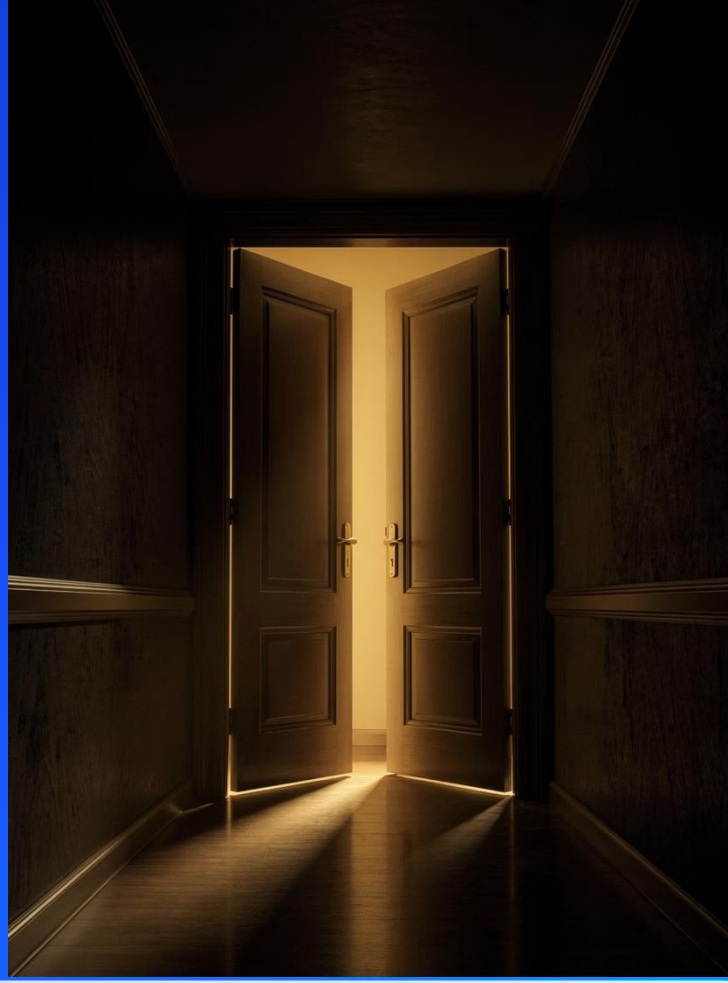


The Call to Adventure

The attacks stopped waiting for the calendar.

Crossing the Threshold

No more once a year.
No more guessing which channel matters.



The Trials

This is where it either holds... or it doesn't.



The Return

Your journey starts where theirs ended.



Implementing HRM – Best Practices



- Engage Leadership
- Adoptive Culture
- Risk Assessment
- Personalize
- Leverage AI
- Monitor
- Incorporate the Human


reddit


OLD HABITS
CHANGE

Everything I learned about information security, I learned...

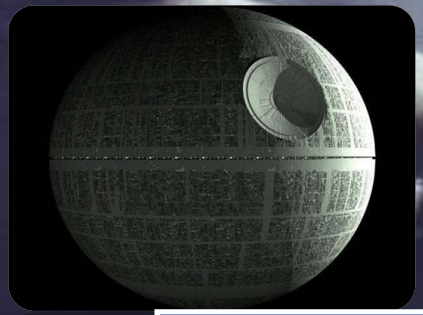



STAR
WARS



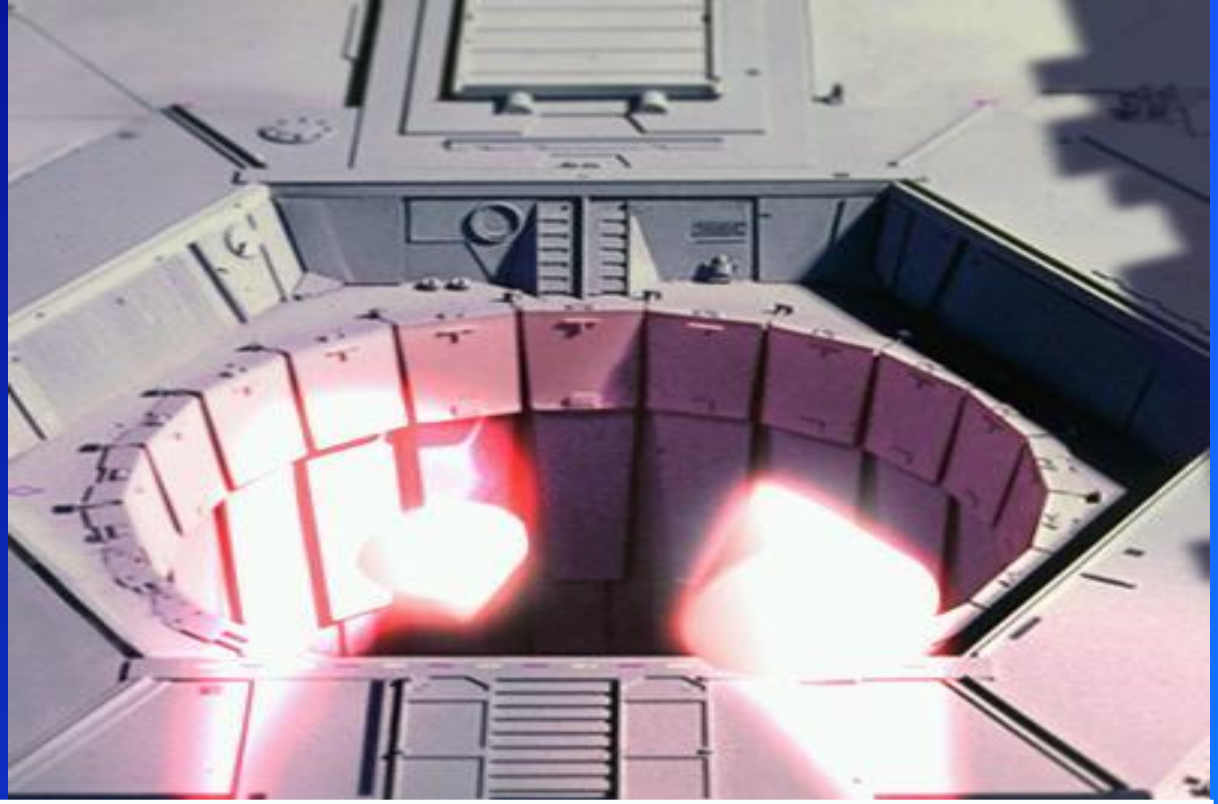
Spoofing


Boring conversation anyway.





Failure to Patch



Vulnerability and exploit leads to destruction



Carbonite – remember to backup!


Social Engineering to gain access

Be careful of those Phishing emails




HACKER!
STAR WARS
R2-D2
STAR WARS WALLPAPER NO. 5 , MADE BY IG-88. STAR WARS: ORD MANTELL HOLONET WWW.STARWARS.PRIV.PL

Be an InfoSec Jedi!





Questions?



Three Things. Next 30 Days

1

Identify Your Stage — Honestly

Assess where your organization genuinely sits on the five-stage model. Use your near-miss report rate as the primary signal. Write it down and share it with one decision-maker.

2

Start Tracking One New Metric

Pick one near-miss metric you are not currently capturing — time-to-report, protocol adherence, repeat click rate — and begin tracking it this week. Data you don't collect can't drive decisions.

3

Find Your Leadership Ally

Identify one leader — a principal, a department head, a superintendent — who will publicly acknowledge the next near-miss report as an organizational win. One visible act reshapes the culture faster than any training module.


Why is Beef Stew, not a good password?
PASSWORD
Because it's not **stroganoff**
ERROR!





Thank You For Your Attention

https://talk.ac/jamesmcquiggan?code=HRM

JAMES MCQUIGGAN

Here’s Your Attendance Slide!

